

Task 1: Build lab Environment

```
link/ether 08:0d:04:76:11:40 brd ff:ff:ff:ff:ff:ff
altname enxe86a64701148
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether 98:3b:8f:ef:7a:4f brd ff:ff:ff:ff:ff:ff
    altname wlx983b8fef7a4f
    inet 192.168.0.61/24 brd 192.168.0.255 scope global dynamic noprefixroute wlp2
s0
        valid_lft 84189sec preferred_lft 84189sec
    inet6 fe80::cd84:8422:b1f7:7ef0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: virbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc htb state UP group def
ault qlen 1000
    link/ether 52:54:00:1b:0a:ec brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.1/24 brd 192.168.100.255 scope global virbr1
        valid_lft forever preferred_lft forever
5: vnet0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master virbr1 s
tate UNKNOWN group default qlen 1000
    link/ether fe:54:00:8a:63:78 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::fc54:ff:fe8a:6378/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

lexie@LexLuthor in ~
└─
```

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Sun Apr 19 05:14:41 2026
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:8a:63:78 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.177/24 brd 192.168.100.255 scope global eth0
        inet6 fe80::5054:ff:fe8a:6378/64 scope link
            valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$
```

Task2: Confirming Connectivity

```
lexie@LexLuthor in ~
└─ ping 192.168.100.177
PING 192.168.100.177 (192.168.100.177) 56(84) bytes of data.
64 bytes from 192.168.100.177: icmp_seq=1 ttl=64 time=0.720 ms
64 bytes from 192.168.100.177: icmp_seq=2 ttl=64 time=0.823 ms
64 bytes from 192.168.100.177: icmp_seq=3 ttl=64 time=0.746 ms
64 bytes from 192.168.100.177: icmp_seq=4 ttl=64 time=0.850 ms
^C
--- 192.168.100.177 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3107ms
rtt min/avg/max/mdev = 0.720/0.784/0.850/0.053 ms

lexie@LexLuthor in ~
└─
```

```
msfadmin@metasploitable:~$ ping -c 2 192.168.100.1
PING 192.168.100.1 (192.168.100.1) 56(84) bytes of data.
64 bytes from 192.168.100.1: icmp_seq=1 ttl=64 time=0.195 ms
64 bytes from 192.168.100.1: icmp_seq=2 ttl=64 time=0.683 ms

--- 192.168.100.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 999ms
rtt min/avg/max/mdev = 0.195/0.439/0.683/0.244 ms
msfadmin@metasploitable:~$
```

Task3: Identifying Live Hosts

```
lexie@LexLuthor in ~  
> nmap -sn 192.168.100.0/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-19 14:46 +0500  
Nmap scan report for 192.168.100.1  
Host is up (0.00027s latency).  
Nmap scan report for 192.168.100.177  
Host is up (0.00083s latency).  
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.11 seconds  
  
lexie@LexLuthor in ~  
> █
```

Task4: Identifying Active ports and Services

```
lexie@LexLuthor in ~  
> nmap -sT 192.168.100.177  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-19 14:47 +0500  
Nmap scan report for 192.168.100.177  
Host is up (0.0022s latency).  
Not shown: 977 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql
```